

# ANSHUL GAWAI

9145064466 | anshulgawai@21gmail.com | linkedin.com/anshul | github.com/Ashking-tech

*AI&DS undergraduate with strong systems and backend engineering background, expanding into ML/DL.*

## EDUCATION

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- **Anantrao Pawar College of Engineering and Research** Pune, Maharashtra  
*Bachelor of Engineering in Artificial Intelligence And Data Science* 2024 - 2028

## PROJECTS

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- **SSH Honeypot | Go, crypto/ssh, ED25519, Docker** 2026
  - Made deception system in Go acting as a legitimate Ubuntu 22.04 SSH server, trapping and profiling malicious actors via a convincing 20+ command interactive shell
  - Built a low-level SSH daemon from scratch using Go's crypto/ssh library, covering ED25519 host key negotiation and geospatial enrichment of captured attack telemetry
  - Containerized the full stack via Docker for one-command deployment; shipped a live monitoring dashboard with aggregated attacker statistics
- **BitTorrent Client | Node.js, TCP/UDP, Binary Protocol** 2025
  - Engineered the BitTorrent wire protocol at the binary level and implemented the full peer-to-peer download pipeline from scratch — UDP tracker negotiation, peer handshaking, choke/unchoke state machines, and concurrent piece scheduling
  - Designed a block-level file reconstruction pipeline with real-time progress tracking; achieved verified end-to-end downloads from live public peers
- **git-visualize | Go, CLI Tooling, Git Internals** 2026
  - A Go CLI utility that recursively scans local repositories and renders a GitHub-style contribution heatmap in the terminal with color-coded commit intensity over a rolling 6-month window
- **Audio Fingerprinting | Go, gonum/fft, WAV/PCM, Signal Processing** 2026
  - Architected a Shazam-style audio recognition engine in Go, implementing a custom WAV decoder for RIFF header parsing and PCM extraction with stereo-to-mono normalization
  - Engineered an FFT-based spectrogram generator using gonum to compute magnitude spectrums and visualize time-frequency data for acoustic peak analysis

## RESEARCH

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- **Adaptive Optics Wavefront Reconstruction Pipeline | C, SIMD, SVD** 2026–Present
  - Developing a deterministic, high-performance C-based pipeline for real-time wavefront reconstruction in adaptive optics, targeting sub-millisecond latency to meet atmospheric coherence time constraints
  - Optimizing sensor processing by implementing Thresholded Center of Gravity (TCoG) with Reciprocal Lookup Tables (LUTs) to eliminate floating-point division bottlenecks and bypass hardware pipeline stalls
  - Engineering an offline Singular Value Decomposition (SVD) truncation method to mathematically eradicate high-frequency waffle-mode errors without incurring real-time spatial filtering overhead

## TECHNICAL SKILLS

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**Languages:** Go, JavaScript, Python, C/C++, Shell/Bash

**Backend & Frameworks:** Node.js, Express.js, REST API Design

**ML & Data:** Python, NumPy, Pandas (learning: PyTorch, Deep Learning)

**Tools:** Git, GitHub, Docker, npm, Nvim, zed, OpenCode, Linux